FIREWALL CONFIGURATION AND SYSTEM HARDENING

Name: Deeksha K

Internship: Cybersecurity Internship – Cognetix Technology

Tool Used: Windows Defender Firewall

Platform: Windows Operating System

INTRODUCTION

Firewall configuration and system hardening are important security practices used to protect computer systems from unauthorized access and network attacks. A firewall acts as a barrier between a trusted internal system and external networks. By properly configuring firewall rules, only legitimate traffic is allowed while harmful traffic is blocked. In this project, Windows Defender Firewall was configured to allow safe ports and block unsafe ports in order to improve system security.

OBJECTIVE

The objective of this project is to configure firewall rules and apply basic system hardening techniques to strengthen system security and reduce the risk of unauthorized access.

TOOLS USED

The tools used in this project include Windows Defender Firewall with Advanced Security and the Windows operating system.

METHODOLOGY

First, Windows Defender Firewall with Advanced Security was opened to view the firewall configuration settings. The firewall status for Domain, Private, and Public profiles was checked to ensure that the firewall was enabled and actively protecting the system.

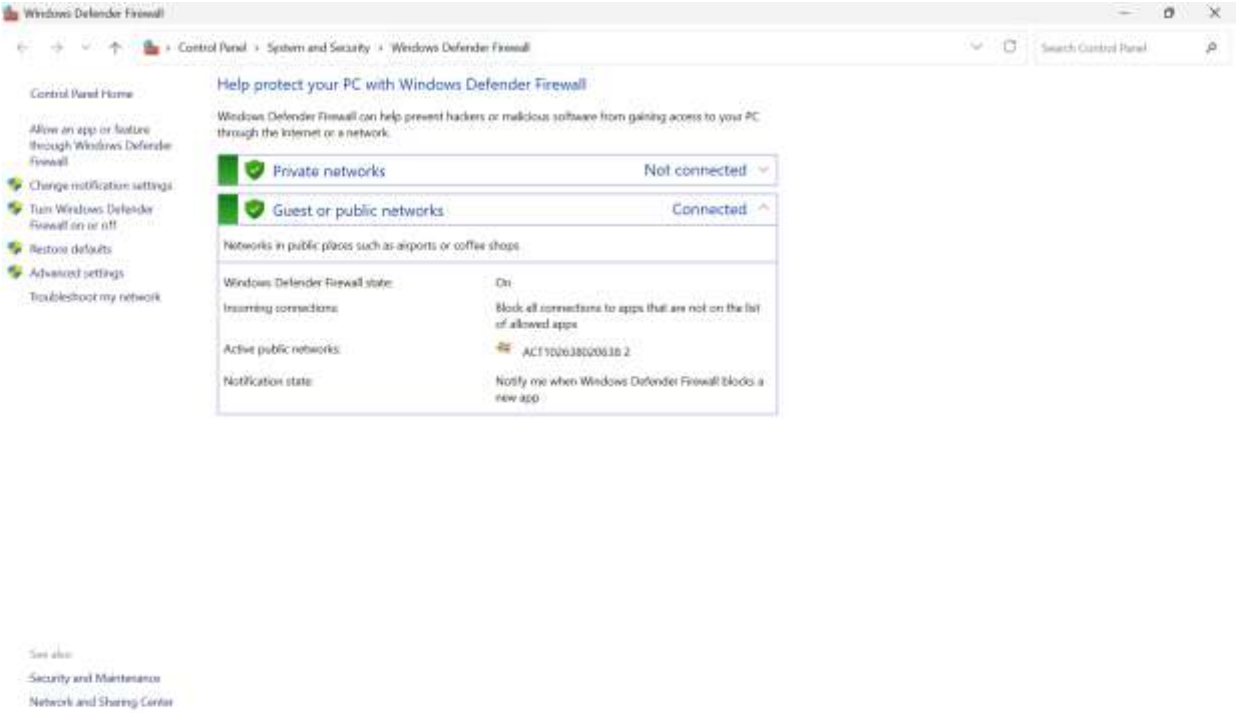
Next, firewall rules were created to allow commonly used safe ports such as port 22, port 80, and port 443. These ports are generally used for secure services like SSH, HTTP, and HTTPS.

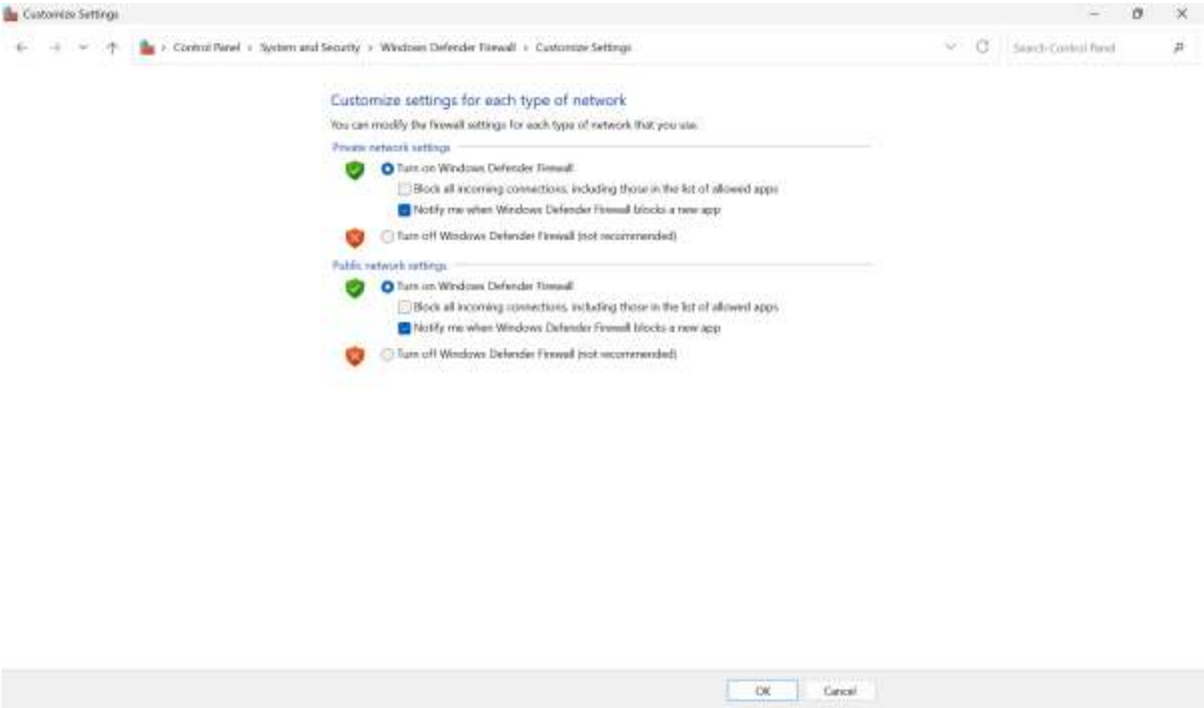
After allowing the safe ports, additional firewall rules were created to block unsafe ports. Ports 21 and 23 were blocked because they are associated with FTP and Telnet services, which are considered insecure and vulnerable to attacks.

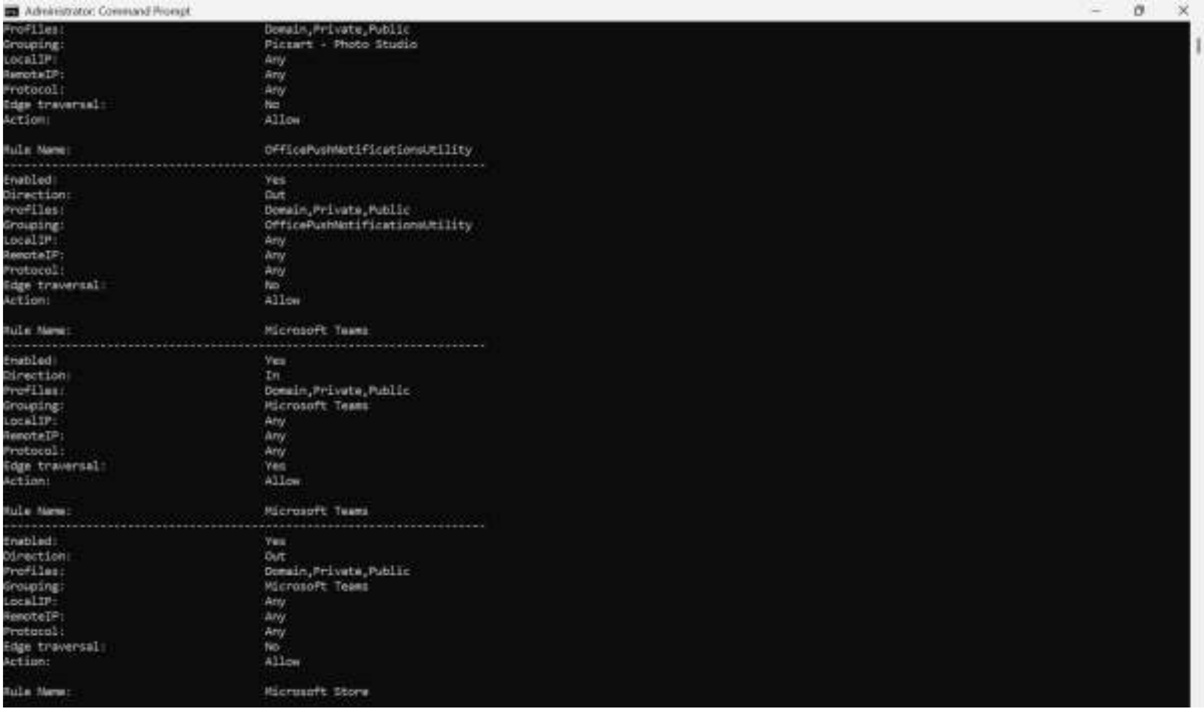
Finally, the firewall rules were reviewed to confirm that the configuration changes were successfully applied.

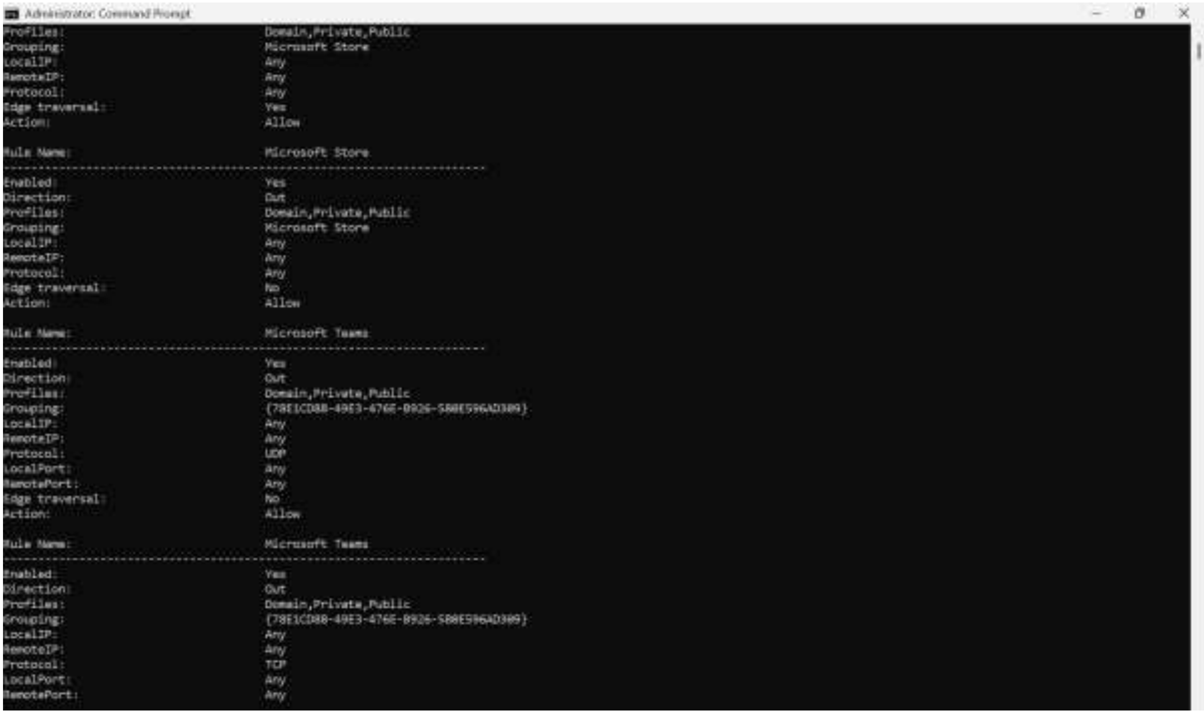
RESULTS

The firewall configuration was successfully implemented. The firewall was active for all network profiles, and rules were created to allow safe ports while blocking unsafe ports. This configuration helps protect the system from unauthorized network access and potential security threats.


Windows Defender Firewall
Control Panel › System and Security › Windows Defender Firewall
Search Control Panel
Control Panel Home
Allow an app or feature through Windows Defender Firewall
Change notification settings
Turn Windows Defender Firewall on or off
Restore defaults
Advanced settings
Troubleshoot my network
Help protect your PC with Windows Defender Firewall
Windows Defender Firewall can help prevent hackers or malicious software from gaining access to your PC through the Internet or a network.
Private networks
Not connected
Guest or public networks
Connected
Networks in public places such as airports or coffee shops
Windows Defender Firewall state:
On
Incoming connections:
Block all connections to apps that are not on the list of allowed apps
Active public networks:
Notification state:
Notify me when Windows Defender Firewall blocks a new app
See also
Security and Maintenance
Network and Sharing Center


Customize Settings
Control Panel › System and Security › Windows Defender Firewall › Customize Settings
Search Control Panel
Customize settings for each type of network
You can modify the firewall settings for each type of network that you use.
Private network settings
Turn on Windows Defender Firewall
Block all incoming connections, including those in the list of allowed apps
Notify me when Windows Defender Firewall blocks a new app
Turn off Windows Defender Firewall (not recommended)
Public network settings
Turn on Windows Defender Firewall
Block all incoming connections, including those in the list of allowed apps
Notify me when Windows Defender Firewall blocks a new app
Turn off Windows Defender Firewall (not recommended)
OK
Cancel





SECURITY ANALYSIS

Proper firewall configuration reduces the attack surface of a system. Allowing only necessary ports ensures that legitimate services can function while preventing attackers from exploiting vulnerable or unnecessary services. Blocking insecure ports such as FTP and Telnet helps strengthen the security of the system.

CONCLUSION

The firewall configuration and system hardening process was successfully completed using Windows Defender Firewall. Safe ports were allowed for essential services, and unsafe ports were blocked to enhance system security. This project demonstrated how firewall rules can be used to protect systems from potential network-based attacks.